Randomness-Recovery Trapdoors: a new methodology for enhancing anamorphic encryption

Xuan Thanh Do * Giuseppe Persiano† Duong Hieu Phan‡ Moti Yung§

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Abstract

The primary goal of Anamorphic encryption (AE), introduced at Eurocrypt 2022, is to enable private communication even in highly adversarial settings, such as when an adversarial *dictator* "legally" confiscates a user's secret keys (compromising the receiver's privacy) and/or coerces users into sending specific messages (compromising the sender's privacy). To achieve this, AE embeds hidden additional messages within seemingly innocuous ciphertexts where the sender and receiver comply with the dictator's demands and, in doing so, AE uncovers novel structural properties of encryption mechanisms. One methodology that extends the capability of a ciphertext is to embed the hidden anamorphic message in the randomness used in encryption. However, not all schemes reveal this randomness as part of the decryption process! Here, we unveil a conceptually simple yet general new methodology that achieves AE. It is based on the concept of *Trapdoor-Aided Randomness Recovery* by which one can generate special key pairs $(\mathsf{pk}, \mathsf{sk})$ that are still indistinguishable from honestly generated key pairs but possess an associated trapdoor td that first allows for randomness extraction from a ciphertext (and nothing more). Secondly, importantly and differently from prior proposals, the new trapdoor should be different from and computationally independent of "the decryption trapdoor key." Primarily, this new methodology allows for a generic construction of *public-key* AE which is a notion introduced at Crypto 24, where, to date, the only known public-key anamorphism relied on a specific CCA encryption scheme. Note that public-key AE eliminates the need for a preliminary private interaction between the receiver and the sender, thus greatly extending the applicability of anamorphism. In addition to obtaining public-key anamorphism, the new methodology, in turn, generically allows for extended anamorphic properties: Specifically and significantly, the methodology allows protections against a dictator that may ask for the randomness employed by the sender.

We then show concrete instantiations of the above methodology based on known lattice-based schemes. Specifically, due to the new methodology, we give efficient anamorphic versions of the Dual Regev scheme and the Lindner-Peikert scheme. Technically, we first define a new problem, called *randomness finding* ($\mathsf{RFinding}$), which requires that even if the adversary obtains the receiver's secret key, then, while it can decrypt, it cannot fully recover the randomness from the ciphertext. Secondly, we reduce the standard LWE assumptions to the hardness of $\mathsf{RFinding}$ for both schemes. Notably, in both schemes we achieve public-key anamorphism utilizing the "trapdoor techniques for lattices" introduced by Micciancio and Peikert at Eurocrypt 2012.

*Institute of Cryptography Science and Technology,Vietnam. thanhkhtn@gmail.com
†Università di Salerno, Italy. giuper@gmail.com
‡Telecom Paris, Institut Polytechnique de Paris, France. hieu.phan@telecom-paris.fr
§Google LLC and Columbia University, USA. motiyung@gmail.com

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1 Introduction

Anamorphic Cryptography [PPY22] was motivated by the need to demonstrate technically and directly that adversarial authorities, which, as part of a crypto-war, demand to obtain the private receiver keys (or to dictate the message to be encrypted by the sender) cannot be successful in abolishing privacy of communication. Specifically, an *Anamorphic* encryption scheme (AE in short) in addition to the regular key generation algorithm KG, has an associated *anamorphic* key generation algorithm aKG that returns the pair $(\mathsf{pk}, \mathsf{sk})$ of public and secret key and a pair $(\mathtt{sdk}, \mathtt{rdk})$ of *double* keys called, respectively, the *sender* and the *receiver double key*. The sender double key $\mathtt{sdk}$ is used to produce an *anamorphic* ciphertext $\mathtt{act}$ that carries two messages, the *regular* message $\mathtt{m}$ and the *anamorphic* message $\mathtt{am}$. The receiver double key $\mathtt{rdk}$ is instead used to decrypt $\mathtt{act}$ and obtain $\mathtt{am}$. The secret key sk can be used to decrypt $\mathtt{act}$ to obtain $\mathtt{m}$ but the anamorphic message $\mathtt{am}$ remains IND-CPA secure even with respect to adversaries possessing sk. The presence of the anamorphic message and key is covert and this is formalized by describing two settings, the *real/regular* setting (in which the keys are generated with KG and each ciphertext carries only the regular message) and the *anamorphic* setting (in which keys are generated with aKG and an anamorphic message is injected into each ciphertext with the use of $\mathtt{sdk}$), and by requiring that they are indistinguishable to an adversary (called the *Dictator*) that has access to the public key pk *and* to the secret key sk. Therefore, the existence of a double receiver key $\mathtt{rdk}$ can be plausibly denied since the regular key generation algorithm KG does not produce such a key. Hence, the anamorphic message $\mathtt{am}$ remains hidden from an overreaching authority (the *Dictator*) that requires access to the receiver's secret information and obtains only the secret key sk.

An important point that should not be overlooked in this scheme is that the double keys are generated by the receiver in conjunction with the public and secret keys. Therefore, there must be a private way for the receiver to send the $\mathtt{sdk}$ to a potential anamorphic sender. In other words, even if we start from a public-key encryption scheme, then its anamorphic mode is private-key. To properly address this discrepancy between the regular and anamorphic schemes (folded into a single ciphertext), the concept of *public-key anamorphism* was introduced in [PPY24], where there is no need for a sender double key, thus dispensing with the need for a private initiation channel between the two parties.

Since its introduction, anamorphism has been proved through three main methodologies: rejection sampling [PPY22, CGM24b], randomness extraction [KPP+23b], and a derivation from the natural redundancy of CCA secure encryption schemes [PPY22, PPY24]. For the notion of public-key anamorphism, things are even more restricted, since the only known public-key anamorphic encryption scheme is Koppula-Waters [KW19] and its anamorphism is proved through its CCA security proof.

In this paper, we introduce a stronger methodology to achieve anamorphism based on extended randomness extraction that we call *Trapdoor-Aided Randomness Extraction*. This is an extension of the randomness extraction methodology used by [KPP+23b] to successfully demonstrate the prevalence of anamorphism by showing that several well-studied/used/standardized encryption schemes, like OAEP-RSA [BR95], El Gamal [ElG85], Cramer-Shoup [CS99] and others, support the anamorphic mode. Essentially, in [KPP+23b], anamorphism was achieved by substituting the randomness used to produce the ciphertext with an encryption of the anamorphic message computed with respect to a symmetric encryption scheme with pseudorandom ciphertexts (e.g., a pseudorandom permutation family), whose encryption key K was part of the sender and of the receiver double key. The receiver extracts the randomness from the ciphertext and then decrypts it with K.

When investigating the structure behind the notion of randomness extractability as considered in [KPP$^+$23b] one notices that access to the decryption key is sufficient. We call this type of randomness extraction *basic* and note two main shortcomings of this notion.

- First, basic extractability is enjoyed by specific encryption schemes as it imposes the minimum requirement for extraction that is compatible with IND-CPA security. Obviously, numerous other schemes do not enjoy this notion, and thus our randomness-trapdoor-aided methodology presented herein will yield *more* schemes which can now become anamorphic.

- Note that in the context of anamorphism, the adversary has access to the decryption key, and therefore basic extractability exposes the randomness to the adversary as well. Therefore, one needs an extra layer of encryption for the anamorphic message to be hidden, and a double key is needed for this, thus making public-key anamorphism difficult to achieve (and this is perhaps the main reason there is only a single such scheme!).

  Conceptually, our trapdoor-aided extraction has the potential to overcome these limitations and thus will yield *stronger* anamorphism (i.e., anamorphism with additional desired properties).

1.1 New Notions and a New Methodology

More technically, we introduce the notion of *Trapdoor Randomness Extractability* and propose this notion in two levels, weaker and stronger, where each level of the new notion allows for new anamorphic constructions for strong forms of anamorphism.

Extractability. Generally speaking, an encryption scheme is *Trapdoor Randomness-Extractable* if it is possible to generate a pair $(\mathsf{pk}, \mathsf{sk})$ of public and secret keys, along with a special (randomness) trapdoor td, and the generated pair is indistinguishable from pairs of keys generated honestly by the usual key generation algorithm (without the special trapdoor). The special trapdoor td can be used in conjunction with sk to extract the randomness from a ciphertext ct. We show that if an encryption scheme is Trapdoor Randomness-Extractable then it enjoys a strong form of an anamorphism that we call *Randomness-Requesting Anamorphism*. Here, the adversary $\mathcal{A}$ obtains the receiver's secret key (as expected in the anamorphic setting for encryption scheme), and, in addition, the adversary can also ask the sender to provide the randomness used to construct each ciphertext.

This demand arises because the adversary (i.e., the *Dictator*) suspects the sender and wants to verify, with absolute certainty, that she encrypted the message exactly as instructed (by combining randomness and the message, then pre-processing and applying the encryption key). Aware that she must provide proof to demonstrate compliance, the sender adopts the following strategy: agrees to provide the randomness $\mathbf{s}$ but before doing so embeds an anamorphic message within $\mathbf{s}$. Specifically, the sender constructs $\mathbf{s}$ as a pseudorandom ciphertext derived from a pre-shared key K (which we assume here and which is agreed upon in advance with the receiver) and the anamorphic message. From the Dictator's point of view, $\mathbf{s}$ is indistinguishable from a truly random value, leaving no room for suspicion and further demands. The sender then uses $\mathbf{s}$ as randomness (with the anamorphic message embedded in it) to encrypt an innocent message. Subsequently, the Dictator is provided with both $\mathbf{s}$ and the resulting ciphertext for verification. The Dictator can independently run the encryption algorithm using the supplied $\mathbf{s}$ and the message he dictated. By comparing the resulting ciphertext with the one provided by the sender, he can confirm that the sender has indeed followed his instructions and sent the intended message. Therefore, in the setting considered by Randomness-Requesting Anamorphism the Dictator has complete knowledge about the ciphertext as it dictates the regular message, it knows the secret key and it knows the randomness used. The Dictator is thus just one step removed from impersonating the sender (i.e., impersonation is when the dictator by himself determines the randomness and the message!).

Finally, the receiver uses the trapdoor td to extract the randomness s hidden within the ciphertext and then uses K to recover the anamorphic message. Note that the trapdoor td is generated along with the pair $(\mathsf{pk}, \mathsf{sk})$ by the receiver itself and therefore needs not to be communicated to the sender.

Strong Extractability. We also consider a stronger notion that we call *Strong* Trapdoor Randomness Extractability in which the special trapdoor td allows for randomness extraction but the same extraction is infeasible for an adversary that does not have access to td. In other words, the encryption function guaranties IND-CPA for the plaintext and one-wayness security for the randomness even with respect to adversaries that, like the Dictator, have access to the secret key.

Under this form of strong extractability, we obtain public-key anamorphism (in the sense of [PPY24]). Public-key anamorphism dispenses with the need of the sender and receiver to have a preliminary exchange/ sharing of the double keys; formally speaking, the sender double key is the empty string. Let us explain how this can be achieved.

If a random oracle H is available and the randomness of a ciphertext is one-way, one can transmit a bit b anamorphically (or a message from a polynomially sized message space) by sampling randomness $\mathbf{s}$ until $H(\mathbf{s}) = b$. Note that the dictator cannot recover all $\mathbf{s}$ (rather it misses a substantial portion thereof) and therefore cannot obtain the anamorphic message b. Also, note that the receiver needs the trapdoor td to extract the randomness from the ciphertext, but the sender needs no extra information and therefore no sender double key must be transferred by the receiver to the sender; in other words, the scheme is *public-key* anamorphic.

The same approach can be used to have an anamorphic KEM: the anamorphic encapsulated message is simply $H(s)$. This gives rise to a hybrid anamorphic scheme.

Contrasting our results with recent impossibility results. The two constructions of anamorphic encryption schemes outlined above consider the underlying encryption schemes as a black box, provided that they are Trapdoor Randomness-Extractable (or Strongly Trapdoor Randomness-Extractable). We stress that our constructions are not in contrast with the upper bound on the message space proved in [CGM24b] for black-box constructions of anamorphic encryption, as we make the additional assumption that the encryption scheme is Trapdoor Randomness-Extractable. We also want to stress that the technique of sampling $\mathbf{s}$ until $H(\mathbf{s}) = b$ is called rejection sampling (this was first used in the context of anamorphism in [PPY22] and before that in [HPRV19] and in [BPR14]). In [CGM25], it is shown that this technique works provided that $\mathbf{s}$ has sufficient min entropy, which is the case since we sample random bits of size $\mathsf{poly}(\lambda)$. This could be a problem if the input to the random oracle of the PRF is a ciphertext in which case [CGM25] showed that anamorphism is not guaranteed if the ciphertext is from a specially designed encryption scheme that has special messages with low min-entropy ciphertexts.

1.2 Applying the new methodology

Having developed our methodology based on trapdoor randomness extraction, we move to applying it to lattice-based encryption schemes, thus obtaining specific anamorphic encryption schemes.

We start by formally realizing the Strong Trapdoor Randomness Extractability for the Dual Regev (DRegev) encryption scheme and the Lindner-Peikert (LP) encryption scheme. To this end, we define a new problem for these lattice-based schemes, which we call RFinding (for randomness finding). This problem requires that even if the adversary obtains the receiver's secret key, while it can decrypt, it cannot fully recover the randomness from the ciphertext. To the best of our knowledge, this problem has not been considered before in lattice-based cryptography. We then reduce the hardness of RFinding to the standard LWE assumption:

- For the DRegev encryption scheme, we surprisingly reduce the k-LWE problem, introduced in [LPSS14] for traitor tracing, to the RFinding problem for the case $k = 1$ (i.e., just one hint which plays the role of the secret key). Since k-LWE is proven to be equivalent to LWE, we demonstrate that RFinding is hard for the DRegev encryption scheme.
- For the LP encryption scheme, we reduce an extended version of LWE, called HNFAextLWE, to the RFinding problem. As shown in [HPS23], the standard LWE reduces to this extended version of LWE. Hence, we prove that RFinding is hard for the LP encryption scheme.

We then show that, by employing the trapdoor techniques introduced by Micciancio and Peikert [MP12], we can embed hidden trapdoors when generating public keys for both the DRegev encryption scheme and the LP encryption scheme. With these trapdoors, the receiver can recover the randomness for both schemes. Along with the hardness of RFinding we achieve Strongly Trapdoor Randomness-Extractable.

We note that since the DRegev scheme is designed to be compatible with the associated trapdoor for IBE, generating a trapdoor does not impose any additional parameter conditions beyond those of the original scheme. In contrast, for the LP encryption scheme, associating the public key with a trapdoor requires a restriction on the size parameters: the public matrix A must have dimensions $n \times m$ with $m \geq n \log q$, which may not be satisfied by all instances of the scheme. The LP scheme is considered the foundational encryption framework for several recent post-quantum cryptosystems standardized by NIST. However, we do not claim the anamorphism for these standards, as their parameters are optimized and do not satisfy our condition. This, instead, opens an interesting question for further research: study anamorphism within NIST standards with their concrete parameters.

1.3 Related work

As we have pointed out, the notion of anamorphic encryption was introduced in [PPY22] and developed further by [KPP+23b] and by [PPY24] that showed deep connections of anamorphism with CCA security. The work of [BGH+24] introduced the notion of robust anamorphic encryption and the notion of anamorphism was extended to other cryptographic primitives in [KPP+23a].

More recently, there has been a flurry of activity around this notion. We mention the foundational work that explore the limitations of anamorphism [CGM24b, CCGM25, DG25, ABG+25] that gives limitations on the efficiency of black-box constructions and encryption schemes that cannot be made anamorphic at all. The work of [BPRS25a] proposed to spread the anamorphic message over several ciphertexts for the purpose of obtaining simpler and more efficient anamorphic encryption schemes. Also, we mention the work of [CCLZ25] that extends the notions of anamorphism and explores the relations among the various notions.

Concurrently to our work, Deo and Libert [DL25] and Banerjee et al. [BPRS25b] have recently and independently proved that the DRegev encryption scheme is public-key anamorphism; they used techniques different from the ones derived from our general methodology, as we outlined above. Indeed, Banerjee et al. [BPRS25b] work in a different kind of multi-message extension which requires more than one ciphertext to recover the anamorphic message (thus losing an important property of zero-latency in encryption). Deo and Libert [DL25] employ an encoding technique to embed the message and then incorporate it into the randomness. This results in a non-black-box construction, followed by a direct security proof of the whole scheme. Our approach is different and more modular. We first show that if the randomness of the ciphertext is one-way even given the secret decryption key, one can construct public-key anamorphic encryption and KEM. Then we apply this general approach to DRegev scheme by establishing the "unrecoverability of randomness" given the secret key (exactly as in the specification of the scheme), from a standard assumption (here, 1-LWE, which is equivalent to LWE, from [LPSS14]). To the best of our knowledge, this is the first work to formally define and achieve this. Our modular approach helps us extend the result to the Lindner-Peikert scheme.

2 Preliminaries

In this section we formally introduce the tools we use in our constructions.

We denote the security parameter by λ. We denote set of real numbers by $\mathbb{R}$ and the integers by $\mathbb{Z}$. Let $[n]$ denote the set $\{1, 2, \ldots, n\}$ for an integer n. For any $q \geq 2$, we let $\mathbb{Z}_q$ denote the ring of integers with addition and multiplication modulo q. Let $\lfloor x \rfloor$ denote the largest integer less than or equal to x, and let the rounding function for a real number x, denoted by $\lfloor x \rceil$, be the closest integer to x. If x is exactly halfway between two integers, the rounding function rounds to the nearest even integer. Denote $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ as the group of reals $[0, 1)$ with mod 1 addition.

A probabilistic polynomial time (PPT) algorithm $\mathcal{A}$ runs in time polynomial in the (implicit) security parameter λ. A function $f(\lambda)$ is negligible in λ if it is $O(\lambda^{-c})$ for every $c \in \mathbb{N}$ and we write $f = \mathsf{negl}(\lambda)$ for short. We say that a probability is overwhelming if it is $1 - \mathsf{negl}(\lambda)$. Similarly, we write $f = \mathsf{poly}(\lambda)$ if $f(\lambda))$ is a polynomial with variable λ. We let $\langle \mathbf{a}, \mathbf{b} \rangle$ denote the inner product of the vectors $\mathbf{a}$ and $\mathbf{b}$ of the same length.

If D is a probability distribution, $x \leftarrow D$ means that x is sampled from D and if S is a countable set, $x \xleftarrow{\$} S$ means that x is sampled uniformly and independently at random from S. We also let $U(S)$ denote the uniform distribution over S. If D_1 and D_2 are distributions over a countable set X, their statistical distance is defined to be $\frac{1}{2} \sum_{x \in X} |D_1(x) - D_2(x)|$ and will be denoted by $\Delta(D_1, D_2)$. We say that two distributions (two ensembles of distributions indexed by n) are statistically close if their statistical distance is negligible in n. Further, we say that two distributions D_0 and D_1 are computationally indistinguishable, denoted $D_0 \approx D_1$, if for all PPT adversaries $\mathcal{A}$, we have

$$|\mathsf{Pr}[x_0 \leftarrow D_0 : \mathcal{A}(1^n, x_0) = 1] - \mathsf{Pr}[x_1 \leftarrow D_1 : \mathcal{A}(1^n, x_1) = 1]| = \mathsf{negl}(n).$$

2.1 Encryption with Pseudorandom Ciphertexts

We recall the notion of a symmetric encryption scheme $\mathsf{pr}\mathbb{E} = (\mathsf{prKG}, \mathsf{prEnc}, \mathsf{prDec})$ with *pseudorandom ciphertexts* using the following game $\mathsf{PRCtG}^b_{\mathsf{pr}\mathbb{E},\mathcal{A}}$, where $b \in \{0,1\}$ and $\mathcal{A}$ is a PPT adversary . We assume that $\mathsf{pr}\mathbb{E}$ for security parameter λ encrypts $n(\lambda)$-bit plaintexts into $\ell(\lambda)$-bit ciphertexts.

$\mathsf{PRCtG}^b_{\mathsf{pr}\mathbb{E},\mathcal{A}}(\lambda)$

1. Set $K \leftarrow \mathsf{prKG}(1^\lambda)$
2. Return $\mathcal{A}^{\mathsf{OPr}^b(K,\cdot)}()$, where

   $\mathsf{OPr}^0(K, \mathtt{m})$ returns a randomly selected $\ell(\lambda)$-bit string;

   $\mathsf{OPr}^1(K, \mathtt{m}) = \mathsf{prEnc}(K, \mathtt{m})$.

Definition 1. *Let* $\mathsf{prE} = (\mathsf{prKG}, \mathsf{prEnc}, \mathsf{prDec})$ *be an IND-CPA symmetric encryption scheme. We say that* prE *has* pseudorandom ciphertexts *if for every PPT adversary* $\mathcal{A}$ *we have*

$$\left|\Pr[\mathsf{PRCtG}^0_{\mathsf{prEnc},\mathcal{A}}(\lambda) = 1] - \Pr[\mathsf{PRCtG}^1_{\mathsf{prEnc},\mathcal{A}}(\lambda) = 1]\right| \leq \mathsf{negl}(\lambda).$$

Note that pseudorandomness of the ciphertexts implies IND-CPA security. Also, symmetric encryption schemes with pseudorandom ciphertexts can be constructed starting from one-way functions.

3 Trapdoor-Aided Randomness Extraction

In this section, we present the notion of an encryption scheme that admits randomness extraction from ciphertexts for specially generated public keys for which a trapdoor td is known. Clearly, the specially generated public keys must be indistinguishable from regular ones and, since we plan to use this notion within the context of anamorphic encryption, this must hold even if the associated secret key is available. We note that, in the context of anamorphism, randomness extraction was first considered by [KPP+23b] but the notion of randomness extraction considered only required knowledge of the secret key and no other trapdoor information was needed. It can be easily seen that the notion introduced in Definition 2 below includes the one of [KPP+23b] as the special case in which td is the empty string.

More formally, we have the following definition.

Definition 2. *Let* $\mathbb{E} = (\mathsf{KG}, \mathsf{Enc}, \mathsf{Dec})$ *be an encryption scheme and let* $r(\cdot)$ *and* $m(\cdot)$ *be two polynomials such that* Enc*, on input a public key with security parameter* λ*, encrypts a* $m(\lambda)$*-bit plaintext using* $r(\lambda)$ *bits. We say that* $\mathbb{E}$ *is* Trapdoor Randomness-Extractable (TRE) *if there exists a pair of efficient algorithms* $(\mathsf{tdKG}, \mathsf{Extr})$ *such that*

1. **Indistinguishability.** tdKG *takes as input security parameter* 1^λ *and returns public key* pk*, secret key* sk*, and trapdoor* td *such that the following two distributions are indistinguishable:*

$$\mathsf{realKG}(\lambda) = \{(\mathsf{pk}, \mathsf{sk}) : (\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KG}(1^\lambda)\}$$

and

$$\mathsf{trapKG}(\lambda) = \{(\mathsf{pk}, \mathsf{sk}) : (\mathsf{pk}, \mathsf{sk}, \mathsf{td}) \leftarrow \mathsf{tdKG}(1^\lambda)\}.$$

2. **Correctness.** Extr *takes as input a ciphertext* ct *and a trapdoor* td *and returns a string* R *such that, for every message* $m \in \{0,1\}^{m(\lambda)}$,

$$\Pr[\, R = r : (\mathsf{pk}, \mathsf{sk}, \mathsf{td}) \leftarrow \mathsf{tdKG}(1^\lambda); r \leftarrow \{0,1\}^{r(\lambda)}; \\ \mathtt{ct} = \mathsf{Enc}(\mathsf{pk}, m; r); R \leftarrow \mathsf{Extr}(\mathtt{ct}, \mathsf{td}) \,] \geq 1 - \mathsf{negl}(\lambda).$$

We note that Property 1 in the definition above implies that if $\mathbb{E} = (\mathsf{KG}, \mathsf{Enc}, \mathsf{Dec})$ is IND-CPA so is $(\mathsf{tdKG}, \mathsf{Enc}, \mathsf{Dec})$. Next we present a stronger notion that essentially says that if the trapdoor td is not known then the randomness cannot be extracted; that is, the ciphertext is, for fixed m, a trapdoor function of the randomness.

Definition 3. *A TRE encryption scheme* $\mathbb{E} = (\mathsf{KG}, \mathsf{Enc}, \mathsf{Dec})$ *is Strongly Trapdoor Randomness-Extractable (sTRE) if for every stateful PPT adversary* $\mathcal{A}$ *there exists a negligible function* negl *such that*

$$\Pr[\mathsf{sTREG}_{\mathbb{E},\mathcal{A}}(\lambda) = 1] \leq \mathsf{negl}(\lambda).$$

where

$\mathsf{sTREG}_{\mathbb{E},\mathcal{A}}(\lambda)$
1. *Set* $(\mathsf{pk}, \mathsf{sk}, \mathsf{td}) \leftarrow \mathsf{tdKG}(1^\lambda)$
2. $\mathtt{m} \leftarrow \mathcal{A}(\mathsf{pk}, \mathsf{sk})$
3. $R \leftarrow \{0,1\}^{r(\lambda)}$
4. $\mathtt{ct} = \mathsf{Enc}(\mathsf{pk}, \mathtt{m}; R)$
5. *Return 1 iff* $\mathcal{A}(\mathsf{pk}, \mathsf{sk}, \mathtt{ct}) = R$

We remark that, rather obviously, if an encryption scheme is sTRE then it is also TRE.

4 Anamorphic Encryption

In this section we review the basic notion of Anamorphic Encryption (AE, see Section 4.1) and then we present a stronger notion of anamorphism for which we give constructions. Specifically, in the basic notion [PPY22] the adversary is given access to the decryption key of the receiver. In Section 4.2 we consider security against an adversary that, in addition to the decryption key of the receiver, also obtains the randomness used by the sender to construct the ciphertext.

For completeness, we review the notions of secure encryption schemes and secure key encapsulation mechanism in Appendix A.

4.1 The Basic Notion

We start by recalling the syntax of an *anamorphic triplet* followed by the notion of receiver anamorphic encryption scheme. Since we want to distinguish between anamorphic sender and anamorphic receiver, we adopt the definition of [CGM24a] further refined by [PPY24]. The original definition of [PPY22] can be obtained by setting the double key `dkey` to be the concatenation of the sender and of the receiver double key (as defined below) and passing `dkey` to the anamorphic encryption and decryption algorithms.

Definition 4 (Anamorphic Triplet)**.** *We say that a triplet* $\mathsf{AME} = (\mathsf{aKG}, \mathsf{aEnc}, \mathsf{aDec})$ *of PPT algorithms is an* anamorphic triplet *if*

- *the* anamorphic key generation *algorithm* aKG *takes as input the security parameter* 1^λ *and returns a pair* $(\mathsf{apk}, \mathsf{ask})$ *of* anamorphic *keys and a pair* $(\mathtt{sdk}, \mathtt{rdk})$ *of* sender *and* receiver double *key;*
- *the* anamorphic encryption *algorithm* aEnc *takes as input the anamorphic public key* apk*, the sender double key* $\mathtt{sdk}$*, and* two *messages, the* regular message $\mathtt{m}$ *and the* anamorphic message $\mathtt{am}$*, and returns an* anamorphic ciphertext $\mathtt{act}$*;*
- *the* anamorphic decryption *algorithm* aDec *takes as input the anamorphic secret key* ask*, the receiver double key* $\mathtt{rdk}$*, and an anamorphic ciphertext* $\mathtt{act}$ *and returns an anamorphic message* $\mathtt{am}$*;*

and, in addition, the following correctness *requirement is satisfied*

- *for every regular message* $\mathtt{m}$ *and anamorphic message* $\mathtt{am}$*, it holds that*

$$\mathsf{aDec}(\mathsf{ask}, \mathtt{rdk}, \mathtt{act}) = \mathtt{am}$$

  except with negligible in λ *probability, where* $((\mathsf{apk}, \mathsf{ask}), (\mathtt{sdk}, \mathtt{rdk})) \leftarrow \mathsf{aKG}(1^\lambda)$ *and* $\mathtt{act} \leftarrow \mathsf{aEnc}(\mathsf{apk}, \mathtt{sdk}, \mathtt{m}, \mathtt{am})$.

We are now ready to present the notion of receiver anamorphic encryption.

Definition 5 (Anamorphic Encryption Scheme)**.** *An IND-CPA secure encryption scheme* $\mathbb{E} = (\mathsf{KG}, \mathsf{Enc}, \mathsf{Dec})$ *is* receiver anamorphic *if there exists an anamorphic triplet* $\mathsf{AME} = (\mathsf{aKG}, \mathsf{aEnc}, \mathsf{aDec})$ *such that for every PPT adversary* $\mathcal{A}$ *there exists a negligible function* negl *such that*

$$\left|\Pr[\mathsf{RealG}_{\mathbb{E},\mathcal{A}}(\lambda) = 1] - \Pr[\mathsf{AnamorphicG}_{\mathsf{AME},\mathcal{A}}(\lambda) = 1]\right| \leq \mathsf{negl}(\lambda),$$

where

$\mathsf{RealG}_{\mathbb{E},\mathcal{A}}(\lambda)$

1. *Set* $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KG}(1^\lambda)$
2. *Return* $\mathcal{A}^{\mathsf{Oe}(\mathsf{pk},\cdot,\cdot)}(\mathsf{pk}, \mathsf{sk})$*, where* $\mathsf{Oe}(\mathsf{pk}, \mathtt{m}, \mathtt{am}) = \mathsf{Enc}(\mathsf{pk}, \mathtt{m})$.

and

$\mathsf{AnamorphicG}_{\mathsf{AME},\mathcal{A}}(\lambda)$

1. *Set* $((\mathsf{apk}, \mathsf{ask}), (\mathtt{sdk}, \mathtt{rdk})) \leftarrow \mathsf{aKG}(1^\lambda)$
2. *Return* $\mathcal{A}^{\mathsf{Oa}(\mathsf{apk},\mathtt{sdk},\cdot,\cdot)}(\mathsf{apk}, \mathsf{ask})$*, where* $\mathsf{Oa}(\mathsf{apk}, \mathtt{sdk}, \mathtt{m}, \mathtt{am}) = \mathsf{aEnc}(\mathsf{apk}, \mathtt{sdk}, \mathtt{m}, \mathtt{am})$.

Finally, we have the following definition of Public-Key Anamorphic Encryption Scheme.

Definition 6 (Public-Key Anamorphic Encryption Schemes)**.** *Let* $\mathbb{E} = (\mathsf{KG}, \mathsf{Enc}, \mathsf{Dec})$ *be an anamorphic encryption scheme with anamorphic triplet* $\mathsf{AME} = (\mathsf{aKG}, \mathsf{aEnc}, \mathsf{aDec})$*. We say that* $\mathbb{E}$ *is* public-key anamorphic *if* aKG *always outputs an empty* $\mathtt{sdk}$*.*

Relation to previous notions. We would like to remark that the notion of Public-Key Anamorphism implies two variations of anamorphism that have been considered in previous work. Specifically, the notion of *Single-Receiver* anamorphism of [KPP+23b] requires security of the regular message with respect to anamorphic senders and it is trivially satisfied by a Public-Key anamorphic scheme. In addition, the notion put forth by [CGM24a] requires that an anamorphic sender cannot use the sender double key $\mathtt{sdk}$ to obtain the regular and anamorphic message from ciphertexts produced by other anamorphic senders. This notion also is trivially implied by Public-Key Anamorphism since an empty $\mathtt{sdk}$ will certainly be of no help.

The random oracle. The defintions in this section are given in the so-called standard model. They can be adapted in a straightforward way to the random oracle model by giving access to all parties (honest algorithms, anamorphic algorithms and advcersary $\mathcal{A}$) to the random oracle. We omit the formal definition.

4.2 Randomness Requesting Adversary

In this section we formalize the concept of *Randomness-Requesting Anamorphism*; that is, anamorphism with respect to an adversary that can request the randomness used to produce a ciphertext. For this, we first augment the notion of an anamorphic triplet by requiring the anamorphic encryption scheme to also output the randomness (so to satisfy the request of the adversary) and then we describe the security game we use to formalize the notion.

We remark that the notion of Randomness-Requesting Anamorphism is especially suited for the case in which the randomness used to construct the ciphertext cannot be extracted with the help of the sk but more information is required. By looking ahead, this setting is captured by our notion of a Strongly Trapdoor Randomness-Extractable encryption scheme (see Definition 3).

Definition 7. *A* randomness-declaring *anamorphic triplet (*$\mathsf{randAME}$*) is an anamorphic triplet* $\mathsf{randAME} = (\mathsf{aKG}, \mathsf{aEnc}, \mathsf{aDec})$ *where* aEnc *has the following extended syntax:*

- *the* anamorphic encryption *algorithm* aEnc *of a* $\mathsf{randAME}$ *takes as input the anamorphic public key* apk*, the sender double key* $\mathtt{sdk}$*, and* two *messages, the* regular message $\mathtt{m}$ *and the* anamorphic message $\mathtt{am}$*, and returns an* anamorphic ciphertext $\mathtt{act}$ *and a string R.*

We are now ready to introduce the concept of a Randomness-Requesting Anamorphism.

Definition 8. *Let* $\mathbb{E} = (\mathsf{KG}, \mathsf{Enc}, \mathsf{Dec})$ *be an IND-CPA secure encryption scheme that uses $r(\lambda)$ random bits to encrypt for security parameter λ. We say that $\mathbb{E}$ is* Randomness-Requesting Anamorphic *if there exists a randomness-declaring triplet* $\mathsf{randAME} = (\mathsf{aKG}, \mathsf{aEnc}, \mathsf{aDec})$ *such that such that for every PPT adversary $\mathcal{A}$ there exists a negligible function* negl *such that*

$$\left|\Pr[\mathsf{RealRandG}_{\mathbb{E},\mathcal{A}}(\lambda) = 1] - \Pr[\mathsf{AnamorphicRandG}_{\mathsf{AME},\mathcal{A}}(\lambda) = 1]\right| \leq \mathsf{negl}(\lambda),$$

where

$\mathsf{RealRandG}_{\mathbb{E},\mathcal{A}}(\lambda)$

1. *Set* $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KG}(1^\lambda)$
2. *Return* $\mathcal{A}^{\mathsf{Oe}(\mathsf{pk},\cdot,\cdot)}(\mathsf{pk}, \mathsf{sk})$, *where* $\mathsf{Oe}(\mathsf{pk}, \mathtt{m}, \mathtt{am})$
   - *randomly select* $R \leftarrow r(\lambda)$
   - *Return* $(\mathsf{Enc}(\mathsf{pk}, \mathtt{m}; R), R)$.

and

$\mathsf{AnamorphicRandG}_{\mathsf{randAME},\mathcal{A}}(\lambda)$

1. *Set* $((\mathsf{apk}, \mathsf{ask}), (\mathtt{sdk}, \mathtt{rdk})) \leftarrow \mathsf{aKG}(1^\lambda)$
2. *Return* $\mathcal{A}^{\mathsf{Oa}(\mathsf{apk},\mathtt{sdk},\cdot,\cdot)}(\mathsf{apk}, \mathsf{ask})$, *where* $\mathsf{Oa}(\mathsf{apk}, \mathtt{sdk}, \mathtt{m}, \mathtt{am}) = \mathsf{aEnc}(\mathsf{apk}, \mathtt{sdk}, \mathtt{m}, \mathtt{am})$.

We stress that in game AnamorphicRandG, algorithm aEnc used by oracle Oa is part of a randomness-declaring triple (see Definition 7) and as such it returns an anamorphic ciphertext act and plausible randomness R for act.

5 Achieving Anamorphism

In this section we describe a general methodology for achieving Anamorphism (called *receiver anamorphism* in the original paper [PPY22]) for any encryption scheme $\mathbb{E}$ that is (Strongly) Trapdoor Randomness-Extractable. We give constructions that leverage the same principle of trapdoor-based randomness extraction and that give different security guarantees.

1. The first construction shows that any TRE encryption scheme is Randomness-Requesting Anamorphic.
2. The second construction shows that any sTRE scheme is public-key anamorphic in the sense of [PPY24] for small message space and in the random oracle model.
3. The third construction gives public-key anamorphic KEM in the random oracle model.

The main idea of all the constructions is to make the trapdoor td part of rdk and then use this to extract the randomness r from the anamorphic ciphertext. In the first construction, the randomness r is actually a ciphertext prct of a symmetric encryption scheme with pseudorandom ciphertext. The sender double key sdk includes the key for encrypting and decrypting prct whereas the receiver double key rdk includes also the trapdoor. Note that even if sdk and sk are available to the adversary $\mathcal{A}$, the randomness r is still indistinguishable from a random string, by sTRE since $\mathcal{A}$ does not have access to td. In the second construction, the randomness is sampled until r such that $H(r) = \mathtt{am}$, where H is a hash function that is specified in the public parameters (i.e., a random oracle). Finally, for the KEM, the value $H(r)$ is the key being encapsulated.

In the next sections, we give more details for the constructions.

5.1 Achieving Randomness-Requesting Anamorphism

In this section we show that if an encryption scheme is Trapdoor Randomness-Extractable then it is Randomness-Requesting Anamorphic. As explained above, the idea is to encrypt the anamorphic message using an encryption scheme with pseudorandom ciphertexts and then to use the ciphertext as randomness.

Theorem 1. *Every Trapdoor Randomness-Extractable encryption scheme is Randomness-Requesting Anamorphic.*

Proof. Consider the following triplet $\mathsf{AME} = (\mathsf{aKG}, \mathsf{aEnc}, \mathsf{aDec})$, constructed by assuming the existence of a Trapdoor Randomness-Extractable encryption scheme $\mathbb{E} = (\mathsf{KG}, \mathsf{Enc}, \mathsf{Dec})$ and of a symmetric encryption scheme with pseudorandom ciphertexts $\mathsf{pr}\mathbb{E} = (\mathsf{prKG}, \mathsf{prEnc}, \mathsf{prDec})$.

1. Algorithm aKG takes as input security parameter 1^λ and runs $\mathsf{tdKG}(1^\lambda)$ to obtain $(\mathsf{pk}, \mathsf{sk}, \mathsf{td})$ and $\mathsf{prKG}(1^\lambda)$ to obtain K. It outputs $(\mathsf{apk}, \mathsf{ask}) := (\mathsf{pk}, \mathsf{sk})$ and $(\mathtt{sdk}, \mathtt{rdk}) := (K, (K, \mathsf{td}))$.

2. Algorithm aEnc takes as input $\mathsf{apk}, \mathtt{sdk}, \mathtt{m}, \mathtt{am}$ and computes the anamorphic ciphertext $\mathtt{act}$ in the following way. First, it sets $R := \mathsf{prEnc}(\mathtt{sdk}, \mathtt{am})$, then it computes $\mathtt{act} = \mathsf{Enc}(\mathsf{apk}, \mathtt{m}; R)$.

3. Algorithm aDec takes as input $\mathtt{act}, \mathsf{apk}, \mathsf{ask}$, and $\mathtt{rdk} = (K, \mathsf{td})$ and runs algorithm Extr on input $\mathtt{act}$, and td to obtain R. Then, it outputs $\mathtt{am} = \mathsf{prDec}(K, R)$.

First of all observe that aDec succeeds except with negligible probability by the correctness property of $(\mathsf{tdKG}, \mathsf{Extr})$ and of $\mathsf{pr}\mathbb{E}$.

Let us now prove that AME is an anamorphic triplet (according to Definition 5) and for this we consider the following hybrid game Hybrid that differs from $\mathsf{AnamorphicG}$ only in the way in which Oa computes its reply. Specifically, instead of encrypting $\mathtt{m}$ using the ciphertext for $\mathtt{am}$ as randomness for Enc, $\mathtt{m}$ is encrypted by running Enc on truly random bits. It is easy to see that by the properties of $\mathsf{pr}\mathbb{E}$ (see Def. 1) Hybrid and $\mathsf{AnamorphicG}$ are indistinguishable.

Let us now look at Hybrid and RealG. Now, the only difference is in the way the pair $(\mathsf{pk}, \mathsf{sk})$ is computed: in Hybrid it is output of tdKG, whereas in RealG it is output by KG. By the Trapdoor Randomness-Extractable property of $\mathbb{E}$, the two games are indistinguishable (see Def. 2).

We finish the proof of the following theorem by observing that, as pointed out before, the existence of a Trapdoor Randomness-Extractable encryption scheme implies the existence of an IND-CPA encryption scheme which in turn implies the existence of one-way functions [IL89]. Finally, it is easy to see that a symmetric encryption scheme with pseudorandom ciphertexts can be constructed under the assumption that one-way functions exist. $\square$

5.2 Achieving Public-Key Anamorphism

In this section we show how to obtain public-key anamorphism (i.e., the $\mathtt{sdk}$ is the empty string) from a Strongly Trapdoor Randomness-Extractable encryption scheme $\mathbb{E} = (\mathsf{KG}, \mathsf{Enc}, \mathsf{Dec})$. We assume the existence of a random oracle H. On input anamorphic message $\mathtt{am}$, the anamorphic encryption algorithms aEnc samples randomness R until $H(R) = \mathtt{am}$ and then uses R to encrypt $\mathtt{m}$ by computing ciphertext $\mathtt{ct} = \mathsf{Enc}(\mathsf{pk}, \mathtt{m}; R)$. Intuitively since $\mathbb{E}$ is Strongly Trapdoor Randomness-Extractable the adversary has no way of computing R and thus $\mathtt{am}$ is hidden.

Theorem 2. *Every Strongly Trapdoor Randomness-Extractable encryption scheme is public-key anamorphic in the random oracle.*

Proof. Consider a Strongly Trapdoor Randomness-Extractable encryption scheme $\mathbb{E} = (\mathsf{KG}, \mathsf{Enc}, \mathsf{Dec})$ with $(\mathsf{tdKG}, \mathsf{Extr})$ as additional algorithms for generating key pairs and trapdoor, and for extracting the randomness. Consider the following anamorphic triplet $(\mathsf{aKG}, \mathsf{aEnc}, \mathsf{aDec})$ with access to a random oracle H:

1. aKG on input λ runs tdKG to obtain $(\mathsf{pk}, \mathsf{sk}, \mathsf{td})$ and sets $\mathtt{apk} := \mathsf{pk}$, $\mathtt{ask} := \mathsf{sk}$, and $\mathtt{rdk} := \mathsf{td}$.

2. aEnc on input $\mathtt{apk}$, $\mathtt{m}$ and $\mathtt{am} \in \{0,1\}$, randomly selects R until $H(R) = \mathtt{am}$ and outputs $\mathtt{act} = \mathsf{aEnc}(\mathsf{apk}, \mathtt{m}; R)$.

3. aDec on input $\mathtt{act}$ and $\mathtt{sdk}$, runs Extr on input $\mathtt{act}$ and $\mathtt{sdk}$ to obtain R and outputs $H(R)$.

First of all observe that by Property 2 of Trapdoor Randomness-Extractable (see Definition 2), aDec succeeds except with negligible probability. Let us now prove that the triplet described above is indeed anamorphic.

We consider the following hybrid game Hybrid that is equal to RealG with the difference that the real encryption oracle Oe is replaced with the anamorphic encryption oracle Oa that, instead of picking random string R to compute $\mathsf{Enc}(\mathsf{pk}, \mathtt{m})$, picks a random string R such that $H(R) = \mathtt{am}$. We claim that the RealG and Hybrid are indistinguishable. Consider a polynomial-time adversary $\mathcal{A}$ and let $p_{\mathcal{A}}$ be the probability that $\mathcal{A}$ queries H for input R. Observe that, conditioned on $\mathcal{A}$ not querying $H(R)$, the view of $\mathcal{A}$ is the same in RealG and Hybrid, therefore the distinguishing probability is at most $p_{\mathcal{A}}$. If $p_{\mathcal{A}} \geq 1/\mathsf{poly}(\lambda)$, for some polynomial poly, then we can use $\mathcal{A}$ to break property Strongly Trapdoor Randomness-Extractable of $\mathbb{E}$. Therefore we can conclude that, if $\mathbb{E}$ is Strongly Trapdoor Randomness-Extractable then RealG and Hybrid are indistinguishable.

Finally, let us now compare Hybrid and $\mathsf{AnamorphicG}$ and observe that the only difference is in the pair $(\mathsf{pk}, \mathsf{sk})$ of keys that, in Hybrid are computed by running KG and in $\mathsf{AnamorphicG}$ are computed by running tdKG. By Property 1 of Trapdoor Randomness-Extractable (see Definition 2) the two distributions are indistinguishable. □

Note that for the anamorphic encryption to be efficient, the anamorphic message space must be of polynomial size.

5.3 Achieving Public-Key Anamorphic KEM

Finally, we describe how to obtain Anamorphic Key-Encapsulation Mechanism (KEM) from a Strongly Trapdoor Randomness-Extractable. In a KEM (see Appendix A for the definition of a KEM and its security properties) the anamorphic message is simply a random message that can be used as a key. In other words the anamorphic encryption scheme only takes one message, the regular message, and returns two objects: a randomly selected key K; and an anamorphic ciphertext $\mathtt{act}$ that carries the regular message and the key K that constitutes the anamorphic message (and can thus be extracted only with the receiver double key $\mathtt{rdk}$). A *Public-Key* Anamorphic KEM can be used to setup an anamorphic encryption scheme (not necessarily a public-key one) without interaction between sender and receiver, just as KEM is used in regular (i.e., non-anamorphic) encryption to bootstrap a private-key encryption scheme. Specifically, the public-key Anamorphic KEM is used by the sender and the receiver to anamorphically establish a common randomness (the key K) that can then be used to generate the $\mathtt{sdk}$ and $\mathtt{rdk}$ for the (non-public key) anamorphic encryption scheme.

Let us now be more formal and define the concept of an Anamorphic Triplet for a KEM.

Definition 9 (Anamorphic KEM Triplet)**.** *We say that a triplet* $\mathsf{AME} = (\mathsf{aKG}, \mathsf{aEnc}, \mathsf{aDec})$ *of PPT algorithms is an* anamorphic KEM triplet *if*

- *the* anamorphic key generation *algorithm* aKG *takes as input the security parameter* 1^λ *and returns a pair* $(\mathsf{apk}, \mathsf{ask})$ *of* anamorphic *keys and a pair* $(\mathtt{sdk}, \mathtt{rdk})$ *of* sender *and* receiver double *key;*

- *the* anamorphic encryption *algorithm* aEnc *takes as input the anamorphic public key* apk*, the sender double key* $\mathtt{sdk}$*, and the* regular message $\mathtt{m}$ *and returns an* anamorphic ciphertext $\mathtt{act}$ *and the anamorphic key* K*;*

- *the* anamorphic decryption *algorithm* aDec *takes as input the anamorphic secret key* ask*, the receiver double key* $\mathtt{rdk}$*, and an anamorphic ciphertext* $\mathtt{act}$ *and returns the anamorphic key* K*;*

and, in addition, the following correctness *requirement is satisfied*

- *for every regular message* $\mathtt{m}$*, it holds that*

$$\mathsf{aDec}(\mathsf{ask}, \mathtt{rdk}, \mathtt{act}) = \mathsf{K}$$

except with negligible in λ *probability, where* $((\mathsf{apk}, \mathsf{ask}), (\mathtt{sdk}, \mathtt{rdk})) \leftarrow \mathsf{aKG}(1^\lambda)$ *and* $(\mathtt{act}, \mathsf{K}) \leftarrow \mathsf{aEnc}(\mathsf{apk}, \mathtt{sdk}, \mathtt{m})$.

We are now ready to define Anamorphic KEM.

Definition 10 (KEM Anamorphic Encryption Scheme)**.** *An IND-CPA secure encryption scheme* $\mathbb{E} = (\mathsf{KG}, \mathsf{Enc}, \mathsf{Dec})$ *is* KEM anamorphic *if there exists an anamorphic KEM triplet* $\mathsf{AME} = (\mathsf{aKG}, \mathsf{aEnc}, \mathsf{aDec})$ *such that for every PPT adversary* $\mathcal{A}$ *there exists a negligible function* negl *such that*

$$\left|\Pr[\mathsf{RealG}_{\mathbb{E},\mathcal{A}}(\lambda) = 1] - \Pr[\mathsf{AnamorphicG}_{\mathsf{AME},\mathcal{A}}(\lambda) = 1]\right| \leq \mathsf{negl}(\lambda),$$

where

$\mathsf{RealG}_{\mathbb{E},\mathcal{A}}(\lambda)$

1. *Set* $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KG}(1^\lambda)$
2. *Return* $\mathcal{A}^{\mathsf{Oe}(\mathsf{pk},\cdot)}(\mathsf{pk}, \mathsf{sk})$*, where*
   $\mathsf{Oe}(\mathsf{pk}, \mathtt{m}) = \mathtt{ct}$
   and $\mathtt{ct} \leftarrow \mathsf{Enc}(\mathsf{pk}, \mathtt{m})$.

and

$\mathsf{AnamorphicG}_{\mathsf{AME},\mathcal{A}}(\lambda)$

1. *Set* $((\mathsf{apk}, \mathsf{ask}), (\mathtt{sdk}, \mathtt{rdk})) \leftarrow \mathsf{aKG}(1^\lambda)$
2. *Return* $\mathcal{A}^{\mathsf{Oa}(\mathsf{apk},\mathsf{sdk},\cdot)}(\mathsf{apk}, \mathsf{ask})$*, where*
   $\mathsf{Oa}(\mathsf{apk}, \mathtt{sdk}, \mathtt{m}) = \mathtt{act}$
   and $(\mathtt{act}, \mathsf{K}) \leftarrow \mathsf{aEnc}(\mathsf{apk}, \mathtt{sdk}, \mathtt{m})$.

A scheme is public-key *KEM anamorphic if, in addition to the above, the* aKG *algorithm always outputs an empty sender double key* $\mathtt{sdk}$.

Remark. We note that in definition above we have two different cryptographic primitives, an encryption scheme and a key encapsulation mechanism, and one behaves as the *host* primitive and the other as the *guest* primitive. More specifically, in Definition 10 above, the host is a public key encryption schemes and the KEM is the guest primitive as a KEM plaintext (the key K) is carried along an encryption plaintext (the regular message $\mathtt{m}$) as part of of the anamorphic ciphertext $\mathtt{act}$.

The random oracle. Definition 9 and 10 are in the standard model but can be modified to work in the random oracle model in straightforward way by giving all parties (the encryption algorithms, the triplet algorithms, and the adversary $\mathcal{A}$) access to the random oracle.

Constructing anamorphic KEM. Constructing a KEM anamorphic triplet for a Strongly Trapdoor Randomness-Extractable encryption scheme is very similar to the construction in the previous section. Specifically, the key K is simply $H(R)$, where H is the random oracle and the R is the randomness used to construct the ciphertext. The anamorphic decryption algorithm aDec uses the trapdoor to extract R and then applies the random oracle H to obtain K.

Theorem 3. *Every Strongly Trapdoor Randomness-Extractable encryption scheme is public-key KEM anamorphic in the random oracle.*

6 The Dual Regev Encryption Scheme

In this section we apply our methodology based on Trapdoor Randomness-Extractable encryption schemes to the Dual Regev public-key encryption scheme (DRegev in short). We proceed in four steps.

First, in Section 6.1, we present the needed background on lattices and on the LWE problem. In Section 6.2, we give a quick description of the DRegev scheme. In Section 6.3 we prove that the DRegev encryption scheme is trapdoor randomness-extractable. Therefore by Theorem 1, we obtain that DRegev is Randomness-Requesting Anamorphic. In Section 6.4 we prove that that the DRegev encryption scheme is strongly trapdoor randomness-extractable. Therefore by Theorem 2, we obtain that DRegev is public-key anamorphic.

6.1 Lattice and k-LWE problem

For two matrices $\mathbf{A}, \mathbf{B}$ of compatible dimensions, let $(\mathbf{A}\|\mathbf{B})$ (or, sometimes, $\left(\frac{\mathbf{A}}{\mathbf{B}}\right)$) denote vertical concatenations of $\mathbf{A}$ and $\mathbf{B}$. For $\mathbf{x} \in \mathbb{Z}^m$, define $\mathbf{x}^+ = (1\|\mathbf{x}) \in \mathbb{Z}^{m+1}$. For $\mathbf{A} \in \mathbb{Z}_q^{m\times n}$, define $\mathrm{Im}(\mathbf{A}) = \{\mathbf{As} \mid \mathbf{s} \in \mathbb{Z}_q^n\} \subseteq \mathbb{Z}_q^m$. For $\mathbf{X} \subseteq \mathbb{Z}_q^m$, let $\mathrm{Span}(\mathbf{X})$ denote the set of all linear combinations of elements of $\mathbf{X}$ and define $\mathbf{X}^\perp$ to be $\{\mathbf{b} \in \mathbb{Z}_q^m \mid \forall \mathbf{c} \in \mathbf{X}, \langle \mathbf{b}, \mathbf{c}\rangle = 0\}$.

Let $\mathbf{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n\} \subset \mathbb{R}^n$ consists of n linearly independent vectors. The n-dimensional lattice Λ generated by the basis $\mathbf{B}$ is $\Lambda = L(\mathbf{B}) = \{\mathbf{Bc} = \sum_{i\in[n]} c_i \cdot \mathbf{b}_i \mid \mathbf{c} \in \mathbb{Z}^n\}$. The length of a matrix $\mathbf{B}$ is defined as the norm of its longest column: $\|\mathbf{B}\| = \max_{1\le i\le n} \|\mathbf{b}_i\|$. Here we view a matrix as simply the set of its column vectors.

For a lattice $L \subseteq \mathbb{R}^m$ and an invertible matrix $\mathbf{S} \in \mathbb{R}^{m\times m}$, we define the Gaussian distribution of parameters L and $\mathbf{S}$ by $D_{L,\mathbf{S}}(\mathbf{b}) = \exp(-\pi\|\mathbf{S}^{-1}\mathbf{b}\|^2)$ for all $\mathbf{b} \in L$.

The q-ary lattice associated with a matrix $\mathbf{A} \in \mathbb{Z}_q^{m\times n}$ is defined as $\Lambda^\perp(\mathbf{A}) = \{\mathbf{x} \in \mathbb{Z}^m \mid \mathbf{x}^t \cdot \mathbf{A} = \mathbf{0} \bmod q\}$. It has dimension m, and a basis can be computed in polynomial-time from $\mathbf{A}$. For $\mathbf{u} \in \mathbb{Z}_q^n$, we define $\Lambda_{\mathbf{u}}^\perp(\mathbf{A})$ as the coset $\{\mathbf{x} \in \mathbb{Z}^m \mid \mathbf{x}^t \cdot \mathbf{A} = \mathbf{u}^t \bmod q\}$ of $\Lambda^\perp(\mathbf{A})$.

Let ν_α denote the 1-dimensional Gaussian distribution with standard deviation α, $\alpha \in (0, 1)$.

Lemma 1 (Theorem 3.1, [AP11]). *There is a probabilistic polynomial-time algorithm that, on input positive integers $n, m, q \geq 2$, outputs two matrices $\mathbf{A} \in \mathbb{Z}_q^{m\times n}$ and $\mathbf{T} \in \mathbb{Z}^{m\times m}$ such that the distribution of $\mathbf{A}$ is within statistical distance $2^{-\Omega(n)}$ from $U(\mathbb{Z}_q^{m\times n})$; the rows of $\mathbf{T}$ form a basis of $\Lambda^\perp(\mathbf{A})$; each row of $\mathbf{T}$ has norm $\leq 3mq^{n/m}$.*

Lemma 2 (GPV, [GPV08, LPSS14]). *Given $\mathbf{A} \in \mathbb{Z}_q^{m\times n}$, an invertible $\mathbf{S} \in \mathbb{R}^{m\times m}$, and $\mathbf{u} \in \mathbb{Z}_q^n$, there exists a probabilistic polynomial-time algorithm that, on input a trapdoor matrix $\mathbf{T} \in \mathbb{Z}^{m\times m}$ for the lattice $\Lambda^\perp(\mathbf{A})$, it outputs a sample from a distribution that is statistically close to $D_{\Lambda_{\mathbf{u}}^\perp(\mathbf{A}),\mathbf{S}}$.*

Definition 11 (The LWE problem). *Let $\alpha > 0$, n and $m = m(n)$ be integers, $q = q(n)$ a prime. The $\mathsf{LWE}_{\alpha,q,m}(n)$ problem is defined as follows. Given $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{m\times n})$, distinguish between the following two distributions over $\mathbb{T}^m$*

$$\frac{1}{q} \cdot U\Big(\mathrm{Im}(\mathbf{A})\Big) + \nu_\alpha^m \quad \textit{and} \quad \frac{1}{q} \cdot U\Big(\mathbb{Z}_q^m\Big) + \nu_\alpha^m,$$

where ν_α denotes the one-dimensional Gaussian distribution with standard deviation α.

The $\mathsf{LWE}_{\alpha,q,m}$ problem is conjectured to be hard for any $\alpha > 0$ and prime q such that $\alpha \cdot q \geq n$, and for any $m = \mathsf{poly}(n)$. Next we introduce a new problem whose hardness implies the hardness of the LWE problem.

Definition 12 (k-LWE problem, [LPSS14]). *Let $k \leq m, \alpha > 0$, and $\mathbf{S} \in \mathbb{R}^{m\times m}$ be an invertible matrix. We denote $\mathbb{T}^{m+1} = (\mathbb{R}/\mathbb{Z})^{m+1}$. The $(k, \mathbf{S})$-LWE problem is: given $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{m\times n})$, $\mathbf{u} \leftarrow U(\mathbb{Z}_q^n)$ and $\mathbf{x}_i \leftarrow D_{\Lambda_{-\mathbf{u}}^\perp(\mathbf{A}),\mathbf{S}}$ for $i \leq k \leq m$, the goal is to distinguish between the distributions (over $\mathbb{T}^{m+1}$)*

$$\frac{1}{q} \cdot U\Big(\mathrm{Im}\Big(\frac{\mathbf{u}^t}{\mathbf{A}}\Big)\Big) + \nu_\alpha^{m+1} \quad \textit{and} \quad \frac{1}{q} \cdot U\Big(\mathrm{Span}_{i\leq k}(1\|\mathbf{x}_i)^\perp\Big) + \nu_\alpha^{m+1},$$

where ν_α denotes the one-dimensional Gaussian distribution with standard deviation α.

In [LPSS14], it was shown that this problem can be polynomially reduced to LWE problem for a specific class of diagonal matrices $\mathbf{S}$. In our work, we only need an arbitrary $\mathbf{S}$ such that $(k, \mathbf{S})$-LWE is hard, and thus the use of $\mathbf{S}$ is implicit. For simplicity in the exposition, we use k-LWE instead of $(k, \mathbf{S})$-LWE.

6.2 The scheme

In this section we briefly review the DRegev encryption scheme, the 1-bit public key encryption scheme from [GPV08] whose security relies on the hardness of the LWE problem.

$\mathsf{DRegev}_{m,q,r,\alpha}$ is parameterized by two integers, m and q, a positive real number r and a 1-dimensional Gaussian distribution ν_α with standard deviation α. $\mathsf{DRegev}_{m,q,r,\alpha}$ is conjectured to be IND-CPA for the following setting of the parameters suggested by [GPV08]: $q \geq 2$ is prime such that $r \geq \omega(\sqrt{\log m}), q \geq 5r(m+1)$, $\alpha \leq 1/(r\sqrt{m+1} \cdot \omega(\sqrt{\log n}))$, $m \geq 2n \log q$.

Next we describe the algorithms of $\mathsf{DRegev}_{m,q,r,\alpha}$ and, from now on, we will drop the explicit mention of the parameters.

$\mathsf{KG}(1^\lambda)$**:** Set $n = \lambda$, randomly select matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ and error vector $\mathbf{e} \leftarrow D_{\mathbb{Z}^m,r}$. The secret key is $\mathsf{sk} = \mathbf{e}$ and the public key is the pair $\mathsf{pk} = (\mathbf{A}, \mathbf{u})$ where $\mathbf{u} = f_{\mathbf{A}}(\mathbf{e}) := \mathbf{Ae} \mod q$ is the syndrome.

$\mathsf{Enc}(\mathsf{pk}, b \in \{0,1\})$**:** Write pk as $\mathsf{pk} = (\mathbf{A}, \mathbf{u})$. Sample $\mathbf{r} \xleftarrow{\$} U(\mathbb{Z}_q^n)$ and $\mathbf{x} \leftarrow \nu_\alpha^m$. Compute $\mathbf{p}^t = \mathbf{r}^t\mathbf{A} + \mathbf{x}^t \in \mathbb{Z}_q^m$, and $c = \mathbf{u}^t\mathbf{r} + x + b\lfloor \frac{q}{2} \rfloor \in \mathbb{Z}_q$. Finally, output ciphertext $\mathtt{ct} = (\mathbf{p}; c) \in \mathbb{Z}_q^m \times \mathbb{Z}_q$.

$\mathsf{Dec}(\mathtt{ct}, \mathsf{sk})$**:** Write $\mathtt{ct}$ as $\mathtt{ct} = (\mathbf{p}, c)$ and sk as $\mathbf{e}$. Compute $z = c - \mathbf{e}^t\mathbf{p} \in \mathbb{Z}_q$. Output 0, if $z \mod q$ is closer to 0 than to $\lfloor \frac{q}{2} \rfloor$; otherwise output 1.

6.3 Dual Regev is Trapdoor Randomness-Extractable

In this section we prove that DRegev is Trapdoor Randomness-Extractable. Therefore, we can apply Theorem 1 and prove that DRegev is randomness-revealing anamorphic. We base our proof on the Micciancio-Peikert method [MP12] of generating a random matrix $\mathbf{A}$ along with a trapdoor matrix $\mathbf{T}$. We have the following definition.

Definition 13. *Fix a* gadget *matrix* $\mathbf{G} \in \mathbb{Z}_q^{n\times nk}$ *and let* $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$. *We say that a short matrix* $\mathbf{T} \in \mathbb{Z}^{m\times nk}$ *is a* trapdoor *for* $\mathbf{A}$ *if*

$$\mathbf{AT} = \mathbf{HG} \bmod q$$

for some invertible matrix $\mathbf{H} \in Z_q^{n\times n}$.

We start by reviewing the Micciancio-Peikert method [MP12] for sampling matrices along with their trapdoors with respect to the gadget matrix $\mathbf{G}$ defined as follows. Consider the gadget vector $\mathbf{g}$

$$\mathbf{g}^t := \begin{bmatrix} 1 & 2 & 2^2 & \cdots & 2^{k-1} \end{bmatrix},$$

where $k = \lceil \log q \rceil$, denote $\mathbf{I}_n$ the $n \times n$ identity matrix and define the matrix gadget

$$\mathbf{G} = \mathbf{I}_n \otimes \mathbf{g}^t = \begin{bmatrix} \mathbf{g}^t & & & \\ & \mathbf{g}^t & & \\ & & \ddots & \\ & & & \mathbf{g}^t \end{bmatrix} \in \mathbb{Z}_q^{n\times nk}.$$

The following algorithm GenTrap samples a random matrix $\mathbf{A}$ along with its trapdoor.

1. Sample $\mathbf{A}_1 \leftarrow U(\mathbb{Z}_q^{n\times m_1})$ and $\mathbf{T} \in \mathbb{Z}_q^{m_1\times nk}$
2. Set $m = m_1 + nk$.
3. Define $\mathbf{A} = [\mathbf{A_1} \| \ \mathbf{A_1T} + \mathbf{G}] \in \mathbb{Z}_q^{n\times m}$.

By applying Left over hash lemma, we have that: If $\mathbf{A}_1$ is taken from uniform distribution over $\mathbb{Z}_q^{n\times m_1}$, $\mathbf{T}$ is taken form Gaussian distribution over $\mathbb{Z}_q^{m_1\times nk}$, and $m_1 \approx nk$ then

$$(\mathbf{A_1}, \mathbf{A_1T}) \approx (\mathbf{A_1}, \mathbf{V}),$$

where $\mathbf{V} \leftarrow U(\mathbb{Z}_q^{n\times nk})$. Therefore, $\mathbf{A} = [\mathbf{A_1} \| \ \mathbf{A_1T} + \mathbf{G}] \in \mathbb{Z}_q^{n\times m}$ is random.

Next we describe algorithm Invert that, given a random matrix $\mathbf{A}$ along with a trapdoor $\mathbf{T}$ and a vector $\mathbf{b} \in \mathbb{Z}_q^m$, returns vector $\mathbf{s} \in \mathbb{Z}_q^n$ and $\mathbf{e} \in \mathbb{Z}_q^m$ such that $\mathbf{b}^t = \mathbf{s}^t\mathbf{A} + \mathbf{e}^t$.

1. Set $\mathbf{H} = \mathbf{I}_n$. We have
$$\mathbf{A} \cdot \left[\frac{\mathbf{T}}{\mathbf{I}_{nk}}\right] = \mathbf{G}.$$

2. Compute
$$\mathbf{b}^t \cdot \left[\frac{\mathbf{T}}{\mathbf{I}_{nk}}\right] = \mathbf{s}^t \cdot \mathbf{G} + \mathbf{e}^t \cdot \left[\frac{\mathbf{T}}{\mathbf{I}_{nk}}\right] \pmod q.$$

3. Extract $\mathbf{s}$ by using the gadget trapdoor $\mathbf{T}_{\mathbf{G}}$ for $\mathbf{G}$. It works if
$$\mathbf{e}^t \cdot \left[\frac{\mathbf{T}}{\mathbf{I}_{nk}}\right] \in [-q/4, q/4).$$

4. Compute $\mathbf{e}^t = \mathbf{b}^t - \mathbf{s}^t\mathbf{A}$.

5. Return $(\mathbf{s}, \mathbf{e})$.

We summarize the above discussion in the following lemma.

Lemma 3 (adapted from Theorem 5.1 [MP12])**.** *There exist algorithms* $\mathsf{GenTrap}$ *and* Invert *such that*

- $\mathsf{GenTrap}(1^n, 1^m, q)$ *is an efficient randomized algorithm that, given any integers* $n \geq 1$, $q \geq 2$, *and sufficiently large* $m = O(n \log q)$, *outputs a parity-check matrix* $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ *and a trapdoor* $\mathbf{T}$ *such that the distribution of* $\mathbf{A}$ *is* $\mathsf{negl}(n)$*-close to uniform.*

- Invert *is an efficient randomized algorithm that, with overwhelming probability over all random choices, on input* $\mathbf{b}^t = \mathbf{s}^t\mathbf{A} + \mathbf{e}^t$, *where* $\mathbf{s} \in \mathbb{Z}_q^n$ *is arbitrary and either* $\|\mathbf{e}\| < q/O(\sqrt{n \log q})$ *or* $\mathbf{e} \leftarrow D_{\mathbb{Z}^m,\alpha q}$ *for* $1/\alpha \geq \sqrt{n \log q} \cdot \omega(\sqrt{\log n})$, *outputs* $\mathbf{s}$ *and* $\mathbf{e}$.

Theorem 4. *The Dual Regev encryption scheme* DRegev *is Trapdoor Randomness-Extractable.*

Proof. The algorithms tdKG and Extr needed by Definition 2 are derived directly from algorithms $\mathsf{GenTrap}$ and Invert, whose existence is guaranteed by Lemma 3.

Specifically,

- Algorithm tdKG is like the KG algorithm of DRegev with the difference that $\mathbf{A}$ is not chosen at random from $\mathbb{Z}_q^{n\times m}$ but it is computed by running $\mathsf{GenTrap}$ that returns $\mathbf{A}$ along with trapdoor $\mathbf{T}$. The secret key $\mathsf{sk} = \mathbf{e}$ is instead sampled from $D_{\mathbb{Z}^m,r}$ as in the KG algorithm of DRegev. Finally tdKG outputs $\mathsf{pk} := (\mathbf{A}, \mathbf{Ae})$, $\mathsf{sk} := \mathbf{e}$, and $\mathsf{td} = \mathbf{T}$.

- Algorithm Extr instead takes a ciphertext $\mathtt{ct}$, the public key $\mathsf{pk} = (\mathbf{A}, \mathbf{u})$ and the trapdoor $\mathsf{td} = \mathbf{T}$, and runs Invert to obtain $\mathbf{r}$ and $\mathbf{x}$ such that $\mathtt{ct} = \mathbf{r}^t\mathbf{A} + \mathbf{x}$. Finally, Extr outputs $\mathbf{r}$.

The proof of the theorem follows directly from Lemma 3. □

By Theorem 1, we have the following corollary.

Corollary 1. *If the* DRegev *encryption scheme is IND-CPA then it is Randomness-Requesting Anamorphic as well.*

6.4 Dual Regev is Strongly Trapdoor Randomness-Extractable

In this section we show that the problem of extracting the randomness used by compute a ciphertext of the DRegev encryption scheme is hard. Specifically, we will show that it is not easier than $1-\mathsf{LWE}$. We have the following definition.

Definition 14. *The* $\mathsf{RFinding}$ *problem for the* $\mathsf{DRegev}_{m,q,r,\alpha}$ *encryption scheme consists of the following:*

Input: $((\mathbf{A}, \mathbf{u}), \mathbf{e}, (\mathbf{p}, c))$ *such that*

- $\mathbf{A} \leftarrow \mathbb{Z}_q^{n\times m}$*;*
- *random vector* $\mathbf{e} \leftarrow D_{\mathbb{Z}^m,r}$*;*
- $\mathbf{u} = \mathbf{A}\mathbf{e}$
- $\mathbf{p} := \mathbf{r}^t\mathbf{A} + \mathbf{x}^t$ *and* $c = \mathbf{u}^t\mathbf{r} + x + \mathtt{m}\lfloor \frac{q}{2} \rfloor)$*, where*
  - $\mathbf{r} \leftarrow \mathbb{Z}_q^n$*;*
  - $\mathbf{x} \leftarrow \nu_\alpha^m$*;*
  - $x \leftarrow \nu_\alpha$*;*
  - $\mathtt{m} \in \{0, 1\}$*;*

Output: $\mathbf{r}$

Note that in the $\mathsf{RFinding}$ problem we are given a DRegev public key $(\mathbf{A}, \mathbf{u})$, a DRegev secret key $\mathbf{e}$, and a DRegev ciphertext $(\mathbf{p}, c)$ and we are asked to compute the randomness $\mathbf{r}$ used to construct the ciphertext. The following lemma shows that the $\mathsf{RFinding}$ problem is no easier than 1-LWE (the special case of k-LWE for $k = 1$).

Lemma 4. *If the problem 1-*LWE *is hard then* $\mathsf{RFinding}$ *is also hard.*

Proof. We assume that efficient algorithm $\mathcal{D}$ solves the $\mathsf{RFinding}$ problem. We will build efficient algorithm $\mathcal{A}$ that solves 1-LWE problem. Indeed, given $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n\times m}), \mathbf{u} \leftarrow U(\mathbb{Z}_q^n), \mathbf{x}_0 \leftarrow D_{\Lambda^\perp_{-\mathbf{u}}(\mathbf{A})}$, the goal of solver $\mathcal{A}$ is to distinguish between the following distributions over $\mathbb{Z}_q^{m+1}$

$$U\left(\mathsf{Im}(\mathbf{A}^+)\right) + \nu_{\alpha q}^{m+1} \quad \text{and} \quad U\left(\mathsf{Span}(\mathbf{x}_0^+)^\perp\right) + \nu_{\alpha q}^{m+1},$$

where $\mathbf{A}^+ = (\mathbf{u}\|\mathbf{A}) \in \mathbb{Z}_q^{n\times(m+1)}$, $\mathbf{x}_0^+ = (1\|\mathbf{x}_0) \in \mathbb{Z}^{m+1}$.

Since the sample $\mathbf{b}$ is drawn from one of the two distributions defined over $\mathbb{Z}_q^{m+1}$, we parse $\mathbf{b} = (\mathbf{b}_1, b_2) \in (\mathbb{Z}_q^m, \mathbb{Z}_q)$.

The solver $\mathcal{A}$ computes the input for $\mathcal{D}$ as follows:

1. Send $\mathbf{A}, \mathbf{u}$ to $\mathcal{D}$.
2. Set $\mathbf{e} := \mathbf{x}_0$.
3. Set $\mathbf{p} := \mathbf{b}_1$ and $c := b_2$.

It then runs $\mathcal{D}$ on the above inputs (including the public key $(\mathbf{A}, \mathbf{u})$, the secret key $\mathbf{e}$ and the ciphertext $(\mathbf{p}, c)$) to obtain a randomness $\mathbf{r} \in \mathbb{Z}_q^n$. If $\mathbf{r}^t \cdot \mathbf{A}^+ \approx \mathbf{b}$ then $\mathcal{A}$ returns that $\mathbf{b}$ was sampled from the distribution $U\left(\mathsf{Im}(\mathbf{A}^+)\right) + \nu_{\alpha q}^{m+1}$; otherwise $\mathcal{A}$ returns that $\mathbf{b}$ was sampled from $U\left(\mathsf{Span}(\mathbf{x}_0^+)^\perp\right) + \nu_{\alpha q}^{m+1}$. $\mathcal{A}$ succeeds because:

- When $\mathbf{b}$ was sampled from the distribution $U\left(\mathsf{Im}(\mathbf{A}^+)\right) + \nu_{\alpha q}^{m+1}$, then $(\mathbf{p}, c)$ is a valid ciphertext drawn from exactly the same distribution as in the Dual Regev encryption scheme. Because $\mathcal{D}$ can solve RandFinding problem, the returned randomness $\mathbf{r}$ satisfies $\mathbf{r}^t \cdot \mathbf{A}^+ \approx \mathbf{b}$.

- When $\mathbf{b}$ is sampled from $U\left(\mathsf{Span}(\mathbf{x}_0^+)^\perp\right) + \nu_{\alpha q}^{m+1}$, we show that the probability that there exists $\mathbf{r}$ such that $\mathbf{r}^t \cdot \mathbf{A}^+ \approx \mathbf{b}$ is negligible. Indeed, as $\mathbf{x}_0^+ \mathbf{A}^+ = 0 \mod q$, the projection of $\mathsf{Im}(\mathbf{A}^+) \mod q$ on $\mathsf{Span}(\mathbf{x}_0^+)^\perp \mod q$ is an identical mapping, and consequently, the probability that a random point sampled from $U\left(\mathsf{Span}(\mathbf{x}_0^+)^\perp\right)$ is close to a lattice point in $\mathsf{Im}(\mathbf{A}^+) \mod q$ is negligible.

□

The hardness of the RFinding problem under the 1-LWE assumption makes the proof of the following theorem straightforward.

Theorem 5. *If the* DRegev *encryption scheme is IND-CPA then it is Strongly Trapdoor Randomness-Extractable as well.*

By Theorem 2, we have the following corollary.

Corollary 2. *If the* DRegev *encryption scheme is IND-CPA then it is public-key anamorphic in the random oracle model.*

By Theorem 3, we have the following corollary.

Corollary 3. *If the* DRegev *encryption scheme is IND-CPA then it is public-key KEM anamorphic in the random oracle model.*

7 Anamorphic for Lindner-Peikert Framework

In this section we apply our methodology based on Trapdoor Randomness-Extractable encryption schemes to the Lindner-Peikert public-key encryption scheme (LP in short). We proceed in four steps.

First, in Section 7.1, we present the needed Extended LWE problem assumption. In Section 7.2, we give a quick description of the LP scheme. In Section 7.3 we prove that the LP encryption scheme is trapdoor randomness-extractable. Therefore, by Theorem 1, we obtain that LP is Randomness-Requesting Anamorphic. In Section 7.4 we prove that the LP encryption scheme is strongly trapdoor randomness-extractable. Therefore, by Theorem 2, we get that LP is public-key anamorphic.

7.1 Extended LWE Assumption

We start by recalling the Extend Learning With Errors (LWE) assumption. The extended-LWE assumption [HPS23] states that pseudorandomness of an LWE instance $(\mathbf{A}, \mathbf{r}^t\mathbf{A} + \mathbf{e}^t)$ still holds when the adversary is given an additional hint $\mathbf{h}$ computed as a function of $\mathbf{r}^t\mathbf{Z}_0$ and $\mathbf{e}^t\mathbf{Z}_1$ for small matrices $\mathbf{Z}_0, \mathbf{Z}_1$. In [HPS23], it is shown that the standard LWE reduces to this extended version of LWE, called HNFAextLWE, which is described as follows:

Challenger $\mathcal{C}^{\mathsf{HNFAextLWE}}$		Adversary $\mathcal{A}$
$\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n\times m})$ $\beta \leftarrow U(\{0,1\})$		
	$\xrightarrow{\mathbf{A}}$	
		$\mathbf{Z}_0, \mathbf{Z}_1, st \leftarrow \mathcal{A}_1(\mathbf{A})$ s.t. $\|\mathbf{Z}_0\| \leq B; \|\mathbf{Z}_1\| \leq B$
	$\xleftarrow{\mathbf{Z}_0, \mathbf{Z}_1}$	
$\mathbf{r} \hookleftarrow \chi^n, \mathbf{e} \hookleftarrow \chi^m, \mathbf{g} \hookleftarrow \chi^\ell$ $\mathbf{h}^t = \mathbf{r}^t \cdot \mathbf{Z}_0 + \mathbf{e}^t \cdot \mathbf{Z}_1 + \mathbf{g}^t \bmod q$ $\mathbf{b}^t = \mathbf{r}^t \cdot \mathbf{A} + \mathbf{e}^t$; if $\beta = 0$ $\mathbf{b}^t = \mathbf{u}^t \hookleftarrow \mathcal{U}\left(\mathbb{Z}_q^m\right)$; if $\beta = 1$		
	$\xrightarrow{\mathbf{b}, \mathbf{h}}$	
		$\beta' \leftarrow \mathcal{A}_2(\mathbf{A}, \mathbf{b}, \mathbf{Z}, \mathbf{h}, st)$
	$\xleftarrow{\beta' \in \{0,1\}}$	

Figure 1: The decision game for $\mathsf{HNFAextLWE}_{q,n,m,\chi,B}$

Definition 15 (HNFAextLWE, adapted from [HPS23])**.** *Let $\lambda \in \mathbb{N}$ be a security parameter. Let $q = q(\lambda), n = n(\lambda), m = m(\lambda), B = B(\lambda) \in \mathbb{N}$ and χ be an error distribution on $\mathbb{R}^m$. The goal of* $\mathsf{HNFAextLWE}_{q,n,m,\chi,B}$ *for an adversary $\mathcal{A}$ is to distinguish between the case where $\beta = 0$ and $\beta = 1$ in the interactive game depicted in Fig. 1. We define the advantage of $\mathcal{A}$ in the* HNFAextLWE *game as*

$$\begin{aligned} & \mathsf{Adv}^{\mathsf{HNFAextLWE}}(\mathcal{A}) \\ = \; & |\Pr[\mathcal{A}(\mathbf{A}, \mathbf{Ar} + \mathbf{e}, \mathbf{Z}_0, \mathbf{Z}_1, \mathbf{h}) \to 1] - \Pr[\mathcal{A}(\mathbf{A}, \mathbf{u}, \mathbf{Z}_0, \mathbf{Z}_1, \mathbf{h}) \to 1]| \end{aligned}$$

where the elements are distributed as shown in Fig. 1.

7.2 The Lindner-Peikert Framework

In this section, we briefly review the LP encryption scheme, a public-key encryption scheme for bit strings of length ℓ from [LP11], whose security relies on the hardness of the LWE problem. There is also a variant of the LP scheme in the module-LWE setting [KKPP20].

The LP scheme is defined by a set of parameters:

- an integer modulus $q \geq 2$;
- two integers $n, m \geq 1$;
- Gaussian parameters s_k and s_e;
- a message alphabet $\Sigma = \{0, 1\}$ along with a message length $\ell \geq 1$.

In this section, we consider the public matrix $\mathbf{A}$ with size $n \times m$, where $m \leq n \log q$. This condition allows us to generate a random matrix $\mathbf{A}$ along with a trapdoor matrix $\mathbf{T}$ using the Micciancio-Peikert method [MP12], in Section 7.3.

We define two functions.

- $\mathsf{encode}\ \Sigma \to \mathbb{Z}_q$, $\mathsf{encode}(b) := b \cdot \lfloor \frac{q}{2} \rfloor$
- $\mathsf{decode} : \mathbb{Z}_q \to \Sigma$, $\mathsf{decode}(\bar{b}) := 0$ if $\bar{b} \in (-\lfloor \frac{q}{4} \rfloor, \lfloor \frac{q}{4} \rfloor) \subset \mathbb{Z}_q$, and 1 otherwise.

We also extend encode and decode component-wise to vector messages $\mathsf{msg} \in \{0,1\}^\ell$.

Next, we describe the algorithms of $\mathsf{LP}_{m,q,\ell,s_k,s_e}$ and, from now on, we will drop the explicit mention of the parameters.

$\mathsf{LP.KG}(1^\lambda)$.

1. Sample $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{n \times m}$.
2. Sample $\mathbf{R}_1 \hookleftarrow D_{\mathbb{Z},s_k}^{n \times \ell}$ and $\mathbf{R}_2 \hookleftarrow D_{\mathbb{Z},s_k}^{m \times \ell}$, and set
$$\mathbf{P} = \mathbf{R}_1 - \mathbf{A} \cdot \mathbf{R}_2 \in \mathbb{Z}_q^{n \times \ell}.$$
3. Set $\mathsf{sk} = \mathbf{R}_2$, and $\mathsf{pk} = (\mathbf{A}, \mathbf{P})$. In matrix form, the connection between the secret key and the public key is represented as:
$$\begin{bmatrix} \mathbf{A} & \mathbf{P} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{R}_2 \\ \mathbf{I} \end{bmatrix} = \mathbf{R}_1 \bmod q.$$

$\mathsf{LP.Enc}(\mathsf{pk}, \mathsf{msg} \in \{0,1\}^\ell)$.

1. Parse $\mathsf{pk} = (\mathbf{A}, \mathbf{P})$.
2. Sample $\mathbf{e} = (\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) \hookleftarrow D_{\mathbb{Z},s_e}^{n} \times D_{\mathbb{Z},s_e}^{m} \times D_{\mathbb{Z},s_e}^{\ell}$.
3. Let $\mathbf{m} = \mathsf{encode}(\mathsf{msg}) \in \mathbb{Z}_q^\ell$, and compute the ciphertext
$$\mathbf{c}^t = \begin{bmatrix} \mathbf{c}_1^t & \mathbf{c}_2^t \end{bmatrix} = \begin{bmatrix} \mathbf{e}_1^t & \mathbf{e}_2^t & \mathbf{e}_3^t + \mathbf{m}^t \end{bmatrix} \cdot \begin{bmatrix} \mathbf{A} & \mathbf{P} \\ \mathbf{I} & \\ & \mathbf{I} \end{bmatrix} \in \mathbb{Z}_q^{1 \times (m+\ell)}.$$

$\mathsf{LP.Dec}(\mathsf{sk}, \mathsf{ct})$.

1. Parse $\mathsf{sk} = \mathbf{R}_2$, $\mathsf{ct} = \mathbf{c}^t = [\mathbf{c}_1^t, \mathbf{c}_2^t] \in \mathbb{Z}_q^{1 \times (m+\ell)}$

2. Compute

$$\begin{bmatrix} \mathbf{c}_1^t & \mathbf{c}_2^t \end{bmatrix} \cdot \begin{bmatrix} \mathbf{R}_2 \\ \mathbf{I} \end{bmatrix} = \left(\mathbf{e}^t + \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{m}^t \end{bmatrix}\right) \cdot \begin{bmatrix} \mathbf{R}_1 \\ \mathbf{R}_2 \\ \mathbf{I} \end{bmatrix} = \mathbf{e}^t \cdot \mathbf{R} + \mathbf{m}^t,$$

where $\mathbf{R} = \begin{bmatrix} \mathbf{R}_1 \\ \mathbf{R}_2 \\ \mathbf{I} \end{bmatrix}$.

3. Output $\mathsf{msg} = \mathsf{decode}(\mathbf{e}^t \cdot \mathbf{R} + \mathbf{m}^t)$.

7.3 Lindner-Peikert is Trapdoor Randomness-Extractable

In this section we prove that LP is Trapdoor Randomness-Extractable. Therefore, we can apply Theorem 1 and prove that LP is randomness-revealing anamorphic. In this section, we also base our proof on the Micciancio-Peikert method [MP12] of generating a random matrix $\mathbf{A}$ along with a trapdoor matrix $\mathbf{T}$.

Theorem 6. *The Lindner-Peikert encryption scheme* LP *is Trapdoor Randomness-Extractable.*

Proof. The algorithms tdKG and Extr needed by Definition 2 are derived directly from algorithms $\mathsf{GenTrap}$ and Invert, whose existence is guaranteed by Lemma 3. Specifically,

- Algorithm tdKG is like the KG algorithm of LP with the difference that $\mathbf{A}$ is not chosen at random from $\mathbb{Z}_q^{n\times m}$ but it is computed by running $\mathsf{GenTrap}$ that returns $\mathbf{A}$ along with trapdoor $\mathbf{T}$. Sample $\mathbf{R}_1 \hookleftarrow D_{\mathbb{Z},s_k}^{n\times \ell}$ and $\mathbf{R}_2 \hookleftarrow D_{\mathbb{Z},s_k}^{m\times \ell}$ as in the KG algorithm of LP. Finally tdKG outputs $\mathsf{pk} := (\mathbf{A}, \mathbf{P} = \mathbf{R}_1 - \mathbf{A}\cdot\mathbf{R}_2)$, $\mathsf{sk} := \mathbf{R}_2$, and $\mathsf{td} = \mathbf{T}$.

- Algorithm Extr instead takes a ciphertext $\mathsf{ct} = (\mathbf{c}_1, \mathbf{c}_2)$, the public key $\mathsf{pk} = (\mathbf{A}, \mathbf{P})$ and the trapdoor $\mathsf{td} = \mathbf{T}$, and runs Invert to obtain $\mathbf{e}_1$ and $\mathbf{e}_2$ such that $\mathbf{c}_1^t = \mathbf{e}_1^t\mathbf{A} + \mathbf{e}_2$. Finally, Extr outputs $\mathbf{e}_1$.

The proof of the theorem follows directly from Lemma 3. □

By Theorem 6, we have the following corollary.

Corollary 4. *The Lindner-Peikert encryption scheme is Randomness-Requesting Anamorphic.*

7.4 Lindner-Peikert is Strongly Trapdoor Randomness-Extractable

In this section we show that the problem of extracting the randomness used by compute a ciphertext of the LP encryption scheme is hard. Specifically, we will show that it is not easier than $\mathsf{HNFAextLWE}$. We set $s_k = s_e = s$. To simplify notation, in this section, we will be concerned only with discrete Gaussian error distributions $\chi = D_{\mathbb{Z},s}$ and χ^k is simply the product distribution of k independent copies of $D_{\mathbb{Z},s}$. We have the following definition.

Definition 16. *The* RFinding *problem for the* $\mathsf{LP}_{m,q,\ell,s_k,s_e}$ *encryption scheme consists of the following: For a* $\mathbf{r} \hookleftarrow \chi^n$*, let* $(\mathbf{e}_2, \mathbf{e}_3) \hookleftarrow \chi^m \times \chi^\ell$*, and* $\mathbf{m} \in \mathbb{Z}_q^\ell$*. Given as input a random matrix* $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$*,* $\mathbf{R}_1 \hookleftarrow \chi^{n\times\ell}$*,* $\mathbf{R}_2 \hookleftarrow \chi^{m\times\ell}$*, define* $\mathbf{P}, \mathbf{c}_1, \mathbf{c}_2$ *as follows*

$$\begin{aligned}\mathbf{P} &= \mathbf{R}_1 - \mathbf{A}\cdot\mathbf{R}_2 \in \mathbb{Z}_q^{n\times\ell}\\ \mathbf{c}_1^t &= \mathbf{r}^t\mathbf{A} + \mathbf{e}_2^t.\\ \mathbf{c}_2^t &= \mathbf{r}^t\mathbf{P} + \mathbf{e}_3^t + \mathbf{m}^t.\end{aligned}$$

The goal is to find and output the vector $\mathbf{r}$*.*

The following lemma shows that the RandFinding problem is no easier than HNFAextLWE problem.

Lemma 5. *If the problem* HNFAextLWE *is hard then* RFinding *is also hard.*

Proof. We assume that $\mathcal{D}$ solves the RFinding problem. We will build $\mathcal{A}$ solves HNFAextLWE problem. Given $\mathbf{A} \leftarrow \mathbb{Z}_q^{n\times m}$, $\mathbf{u} \leftarrow \mathbb{Z}_q^m$, and $\mathbf{Z}_0 \hookleftarrow \chi^{n\times\ell}$ and $\mathbf{Z}_1 \hookleftarrow \chi^{m\times\ell}$, the goal of solver $\mathcal{A}$ is to distinguish between the following distributions

$$(\mathbf{A},\, \mathbf{r}^t\cdot\mathbf{A} + \mathbf{e}^t,\, \mathbf{Z}_0,\, \mathbf{Z}_1,\, \mathbf{r}^t\cdot\mathbf{Z}_0 + \mathbf{e}^t\cdot\mathbf{Z}_1 + \mathbf{g}^t) \;\text{ and }\; (\mathbf{A},\, \mathbf{u}^t,\, \mathbf{Z}_0,\, \mathbf{Z}_1,\, \mathbf{r}^t\cdot\mathbf{Z}_0 + \mathbf{e}^t\cdot\mathbf{Z}_1 + \mathbf{g}^t),$$

where $\mathbf{r} \hookleftarrow \chi^n$, $\mathbf{e} \leftarrow \chi^m$, $\mathbf{g} \leftarrow \chi^\ell$.

The solver $\mathcal{A}$ computes the inputs for $\mathcal{D}$ as follows:

1. Receive a challenge $\mathbf{b}$ from left or right distribution.
2. Set $\mathbf{R}_1 := \mathbf{Z}_0$, $\mathbf{R}_2 := \mathbf{Z}_1$ and compute $\mathbf{P} = \mathbf{R}_1 - \mathbf{A}\cdot\mathbf{R}_2$.
3. Send $\mathbf{A}, \mathbf{P}, \mathbf{R}_1, \mathbf{R}_2$ to $\mathcal{D}$.
4. For $\mathbf{m} \in \mathbb{Z}_q^\ell$, set $\mathbf{c}_1 := \mathbf{b}$, and simulate $\mathbf{c}_2$ from hints as follows

$$\mathbf{c}_2^t := -\mathbf{c}_1^t\cdot\mathbf{R}_2 + \mathbf{r}^t\cdot\mathbf{R}_1 + \mathbf{e}^t\cdot\mathbf{R}_2 + \mathbf{g}^t + \mathbf{m}^t.$$

If $\mathbf{b}$ comes from the left distribution then $\mathbf{b}^t = \mathbf{r}^t\cdot\mathbf{A} + \mathbf{e}^t$. The adversary $\mathcal{A}$ then runs $\mathcal{D}$ on the above inputs to obtain a randomness $\mathbf{r} \in \mathbb{Z}_q^n$. If $\mathbf{r}^t\cdot\mathbf{A} \approx \mathbf{b}^t$ then $\mathcal{A}$ returns that $\mathbf{b}$ was sampled from the left distribution

$$(\mathbf{A},\, \mathbf{r}^t\cdot\mathbf{A} + \mathbf{e}^t,\, \mathbf{Z}_0,\, \mathbf{Z}_1,\, \mathbf{r}^t\cdot\mathbf{Z}_0 + \mathbf{e}^t\cdot\mathbf{Z}_1 + \mathbf{g}^t);$$

otherwise $\mathcal{A}$ returns that $\mathbf{b}$ was sampled from the right distribution $(\mathbf{A},\, \mathbf{u}^t,\, \mathbf{Z}_0,\, \mathbf{Z}_1,\, \mathbf{r}^t\cdot\mathbf{Z}_0+\mathbf{e}^t\cdot\mathbf{Z}_1+\mathbf{g}^t)$. The adversary $\mathcal{A}$ succeeds because:

- When $\mathbf{b}$ was sampled from the distribution

$$(\mathbf{A},\, \mathbf{r}^t\cdot\mathbf{A} + \mathbf{e}^t,\, \mathbf{Z}_0,\, \mathbf{Z}_1,\, \mathbf{r}^t\cdot\mathbf{Z}_0 + \mathbf{e}^t\cdot\mathbf{Z}_1 + \mathbf{g}^t),$$

$(\mathbf{c}_1, \mathbf{c}_2)$ is a correct ciphertext. Indeed,

$$\begin{aligned}
\mathbf{c}_2^t &= -\mathbf{c}_1^t \cdot \mathbf{R}_2 + \mathbf{r}^t \cdot \mathbf{R}_1 + \mathbf{e}^t \cdot \mathbf{R}_2 + \mathbf{g}^t + \mathbf{m}^t \\
&= -(\mathbf{A} \cdot \mathbf{r} + \mathbf{e})^t \cdot \mathbf{R}_2 + \mathbf{r}^t \cdot \mathbf{R}_1 + \mathbf{e}^t \cdot \mathbf{R}_2 + \mathbf{g}^t + \mathbf{m}^t \\
&= -\mathbf{r}^t \cdot \mathbf{A} \cdot \mathbf{R}_2 - \mathbf{e}^t \cdot \mathbf{R}_2 + \mathbf{r}^t \cdot \mathbf{R}_1 + \mathbf{e}^t \cdot \mathbf{R}_2 + \mathbf{g}^t + \mathbf{m}^t \\
&= \mathbf{r}^t \cdot (\mathbf{R}_1 - \mathbf{A} \cdot \mathbf{R}_2) + \mathbf{g}^t + \mathbf{m}^t \\
&= \mathbf{r}^t \cdot \mathbf{P} + \mathbf{g}^t + \mathbf{m}^t.
\end{aligned}$$

Because $\mathcal{D}$ can solve $\mathsf{RFinding}$ problem, the returned randomness $\mathbf{r}$ satisfies $\mathbf{r}^t \cdot \mathbf{A} \approx \mathbf{b}^t$.

- When $\mathbf{b}$ is sampled from the right distribution

$$(\mathbf{A},\, \mathbf{u}^t,\, \mathbf{Z}_0,\, \mathbf{Z}_1,\, \mathbf{r}^t \cdot \mathbf{Z}_0 + \mathbf{e}^t \cdot \mathbf{Z}_1 + \mathbf{g}^t).$$

  Existence of $\mathbf{r}$ with $\mathbf{r}^t \cdot \mathbf{A} \approx \mathbf{b}^t = \mathbf{u}^t$ implies that $\mathbf{r}^t \cdot \mathbf{A}$ is close to a random vector $\mathbf{u}^t$. This only holds with negligible probability.

$\square$

The hardness of the $\mathsf{RFinding}$ problem under the $\mathsf{HNFAextLWE}$ assumption makes the direct proof of the following theorem straightforward.

Theorem 7. *If the* LP *encryption scheme is IND-CPA then it is Strongly Trapdoor Randomness-Extractable as well.*

By Theorem 2, we have the following corollary.

Corollary 5. *If the* LP *encryption scheme is IND-CPA then it is public-key anamorphic as well.*

By Theorem 3, we have the following corollary.

Corollary 6. *If the* LP *encryption scheme is IND-CPA then it is public-key KEM anamorphic as well.*

8 Conclusions

In this paper we have given a new avenue to construct public-key anamorphic encryption schemes and we have used it to show that two notable lattice-based encryptions schemes, the Dual Regev and the Lindner-Peikert scheme, are public-key anamorphic. Even though the Lindner-Peikert scheme serves as a conceptual basis for recent post-quantum standard, the strict parameter optimization introduced in the standards makes our techniques ineffective. We therefore leave as an open problem to show the anamorphic properties of the post-quantum encryption standards.

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A Public Key Encryption and KEM

We review the standard notions of a public key encryption scheme and of a key encapsulation mechanism and the associated security notions.

Definition 17. *A* public key encryption scheme *is a tuple* $\mathbb{E} = (\mathsf{KG}, \mathsf{Enc}, \mathsf{Dec})$ *of PPT algorithms with the following syntax*

1. *the* key-generator algorithm KG *takes as input the security parameter* 1^λ*, and returns a* public key pk *and a* secret key sk*;*

2. *the* encryption algorithm Enc *takes as input a public key* pk *and a message* m*, and returns a ciphertext* $\mathtt{ct}$*;*

3. *the* decryption algorithm Dec *takes as input a ciphertext* $\mathtt{ct}$ *and a secret key* sk*, and returns a message* m *or the symbol* $\perp$*.*

and that satisfy the following correctness requirement:

- *there exists a negligible function* negl *such that for all* m

$$\Pr\left[\begin{array}{ll}\mathsf{Dec}(\mathsf{sk}, \mathtt{ct}) \neq m : & (\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KG}(1^\lambda) \\ & \mathtt{ct} = \mathsf{Enc}(\mathsf{pk}, m)\end{array}\right] \leq \mathsf{negl}(\lambda).$$

We will consider the $\mathsf{IND-CPA}$ notion of security for $\mathbb{E}$.

Definition 18. *A public key encryption scheme* $\mathbb{E}$ *is called* $\mathsf{IND-CPA}$ secure *if, for every PPT adversary* $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$*, there exists a negligible function* negl *such that*

$$\Pr\left[\begin{array}{ll} & (\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KG}(1^\lambda) \\ & (m_0, m_1, \mathsf{st}) \leftarrow \mathcal{A}_1(\mathsf{pk}) \\ b = b' : & b \xleftarrow{\$} \{0,1\} \\ & \mathtt{ct}^* \leftarrow \mathsf{Enc}(\mathsf{pk}, m_b) \\ & b' \leftarrow \mathcal{A}_2(\mathsf{st}, \mathtt{ct}^\star)\end{array}\right] - \frac{1}{2} \leq \mathsf{negl}(\lambda).$$

Next we define the concept of a key encapsulation mechanism.

Definition 19. *A* key encapsulation mechanism *is a tuple* $\mathsf{KEM} = (\mathsf{KG}, \mathsf{Enc}, \mathsf{Dec})$ *of PPT algorithms with the following syntax*

1. *the* key-generator algorithm KG *takes as input the security parameter* 1^λ*, and returns a* public key pk *and a* secret key sk*;*

2. *the* encapsulation algorithm Enc *takes as input a public key* pk *and returns a ciphertext* $\mathtt{ct}$ *and a key* K*;*

3. *the* decryption algorithm Dec *takes as input a ciphertext* $\mathtt{ct}$ *and a secret key* sk*, and returns a key* K *or the symbol* $\perp$*.*

and that satisfy the following correctness requirement:

- *there exists a negligible function* negl *such that*

$$\Pr\left[\begin{array}{ll}\mathsf{Dec}(\mathsf{sk},\mathtt{ct}) \neq \mathsf{K}: & (\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KG}(1^\lambda) \\ & (\mathtt{ct},\mathsf{K}) = \mathsf{Enc}(\mathsf{pk})\end{array}\right] \leq \mathsf{negl}(\lambda).$$

Following is the formalization of a the concept of a secure KEM.

Definition 20. *A KEM* $\mathsf{KEM} = (\mathsf{KG},\mathsf{Enc},\mathsf{Dec})$ *that, for public key with security parameter* λ *returns* $\ell(\lambda)$*-bit keys, is called* $\mathsf{IND}-\mathsf{CPA}$ secure *if the following two distributions*

$$\mathsf{Real}_{\mathsf{KEM}}(\lambda) = \{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KG}(1^\lambda); (\mathtt{ct},\mathsf{K}) = \mathsf{Enc}(\mathsf{pk}) : (\mathsf{pk},\mathtt{ct},\mathsf{K})\}$$

and

$$\mathsf{Random}_{\mathsf{KEM}}(\lambda) = \{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KG}(1^\lambda); (\mathtt{ct},\mathsf{K}) = \mathsf{Enc}(\mathsf{pk}); \mathsf{K}' \leftarrow \{0,1\}^{\ell(\lambda)} : (\mathsf{pk},\mathtt{ct},\mathsf{K}')\}$$

are computationally indistinguishable.